# Zhonghao Zhang

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## EDUCATION

**Vanderbilt University** Nashville, TN
Bachelor of Engineering in Electrical and Computer Engineering & Mathematics *May 2023 – May 2027*
Minor in Computer Science

- **GPA: 3.85 / 4.0**

- **Relevant Coursework:** Robotics, Control Systems, Applied Machine Learning, Foundations of Machine Learning, Image Processing, Computer Vision, Intermediate Software Design, Digital Signal Processing, Operating Systems, Linear Algebra

## PUBLICATIONS

[1] **Zhang, Z.***, Wu, M.*, Yang, H., Acar, A., Kuntz, A., and Wu, J. Y. "Object-Centered Reconstruction for Vision-Based 3D Force Estimation." Submitted to the *IEEE International Conference on Robotics and Automation* (ICRA), 2027.

[2] Wu, M.*, **Zhang, Z.***, Yang, H., Kuntz, A., and Wu, J. Y. "Transferable Tool-Tissue Contact Detection from Stereo Depth in Robot-Assisted Surgery." Submitted to *SPIE Medical Imaging*, 2027. (*Co-first authors)

[3] **Zhang, Z.**, Zhou, H., Yang, H., Fischer, G. S., and Kazanzides, P. "Dynamics Estimation for the da Vinci Research Kit: A Review of Model-Based and Learning-Based Approaches." *Machines*, vol. 14, no. 8, p. 879, 2026. DOI: 10.3390/machines14080879

[4] **Zhang, Z.**, Lu, M., Lu, Z., Huo, Y., and Yan, X. "Reducing RF-Induced Heating of DBS in 3 T MRI Using Dual-Role Receive Arrays as Wireless Resonators: A Simulation Study." *IEEE Access*, vol. 14, pp. 112236–112247, 2026. DOI: 10.1109/ACCESS.2026.3713428

[5] **Zhang, Z.**, Lu, M., Liang, H., Zu, Z., Gu, Y., Wang, X., Huo, Y., and Yan, X. "Fast Electromagnetic and RF Circuit Co-Simulation for Passive Resonator Field Calculation and Optimization in MRI." *Magnetic Resonance Imaging*, vol. 129, p. 110644, 2026. DOI: 10.1016/j.mri.2026.110644

[6] Yang, H., **Zhang, Z.**, Al-Zogbi, L., Kazanzides, P., and Wu, J. Y. "Friction Compensation for Improved Sensorless Force Estimation in the dVRK-Si." Submitted to *IEEE Transactions on Medical Robotics and Bionics*, 2026.

[7] Zhou, H., Yang, H., Burkhart, B., Deguet, A., **Zhang, Z.**, Fichera, L., Fischer, G. S., Wu, J. Y., and Kazanzides, P. "Towards Dynamic Model Identification and Gravity Compensation for the dVRK-Si Patient Side Manipulator." *IEEE Transactions on Medical Robotics and Bionics*, accepted 2026.

[8] **Zhang, Z.**, Lu, M., Wang, X., Huo, Y., and Yan, X. "Electromagnetic Field Optimization for Safe MRI of Patients with Deep Brain Stimulation Using Wireless Coils." Abstract accepted at the ISMRM & ISMRT Annual Meeting and Exhibition, Cape Town, South Africa, 2026.

[9] **Zhang, Z.,** Wu, M., Yang, H., Acar, A., Kuntz, A., and Wu, J. Y. "Stereo Vision-Based Force Estimation and 3D Reconstruction Toward Autonomous Colorectal Anastomosis: A Preliminary Study." Submitted to the 2nd International Workshop on the Evolving Landscape of Surgical Robotics (ELSR), held in conjunction with the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

## PROFESSIONAL EXPERIENCE

**Machine Automation, Perception and Learning (MAPLE) Lab, Vanderbilt University** Nashville, TN
***Research Assistant, advised by Prof. Jie Ying Wu*** *Jan. 2026 – Present*

- Developed a vision-based force estimation pipeline for the da Vinci Surgical System Si, integrating SAM 2-based stereo reconstruction, Chamfer-loss registration with local linear regularization, and a Set Transformer–LSTM model for force prediction
- Built a vision-based contact detection pipeline using SAM 2 segmentation and FoundationStereo depth estimation, with a two-state hidden Markov model incorporating temporal features for robust contact classification

- Implemented sensorless force estimation on the dVRK-Si by identifying robot kinematics and dynamics parameters for gravity compensation and computed-torque feedforward control
- Developed a C++ stereo video acquisition pipeline with ROS integration for synchronized data collection

**RF Laboratory, Vanderbilt University** Nashville, TN
***Research Assistant, advised by Prof. Xinqiang Yan*** *Dec. 2024 – Feb. 2026*

- Designed 2-, 6-, and 32-element wireless RF coil models in ANSYS HFSS for 3T MRI and developed a fast electromagnetic co-simulation algorithm reducing computation time from hours to milliseconds
- Built a physics-informed deep learning framework in PyTorch integrating EM co-simulation, ResNet-18, and genetic algorithms to optimize RF coil parameters for reducing heating in DBS implants and improving field homogeneity

**Robotics and Autonomous Systems Laboratory, Vanderbilt University** Nashville, TN
***Research Assistant, advised by Prof. Nilanjan Sarkar*** *Aug. 2024 – Dec. 2024*

- Developed an embodied AI assistant for dementia care, using LLMs to generate context-aware non-verbal gestures and driving a 3D avatar in Unity for real-time human-AI interaction, with caregiver feedback integrated to improve emotional appropriateness

**Intelligent Sensor and Detection Technology Laboratory, Shandong University** Shandong, China
***Research Assistant, advised by Prof. Mingshun Jiang*** *May 2024 – Aug. 2024*

- Built a real-time sensor data acquisition and visualization system using Java socket communication over Ethernet, with digital signal processing pipelines for noise reduction and a PyQt GUI for live monitoring and analysis

**Rock robot** Shanghai, China
***Engineer Assistant*** *Apr. 2023 – Jul. 2023*

- Designed and optimized infrared camera mount structures in SolidWorks, and implemented a Python-based computer vision heat source detection pipeline using OpenCV for thermal imaging applications

## TECHNICAL SKILLS

- **Programming Languages:** Python (advanced), MATLAB (advanced), Java (advanced), C++ (advanced)
- **Software and Tools:** ANSYS HFSS, SolidWorks, LTspice